10^5 MPa.

Q.3 Attempt as follows:

Total No. of Pages 3 *Roll No. .....*

FOURTH SEMESTER **B.Tech.(AE/ CE/ EP/ ME/ MC/ PE/ PS)**

END SEMESTER EXAMINATION ***(May -2017)***

HU-202 - ENGINEERING ECONOMICS

Time: 3 Hour ***Max. Marks: 50***

Note: Answer any five questions. Assume suitable missing data, if any.

1.a What do you mean by Fiscal Policy? How it can be used to control environmental degradation? 5

1.b A finance company advertises two investment plans. In plan 1, the company pays Rs. 12,000 after 15 year for every Rs. 1000 invested now. In plan 2, for every Rs. 1000 invested, the company pays Rs. 4000 at the end of the 10th year and Rs. 4000 at the end of 15th year. Select the best investment plan for the investor's point of view at the rate of 12% compounded annually. Use present worth method for calculation. 5

2.a Discuss Business Cycle. Discuss role of engineer in bringing out the economy out of recession. 5

2.b A computer producing company can either purchase the screen or can manufacture within the company. The detail is given below 5

	Purchase	Manufacture
Price per screen	4000	
Tax rate	@ 10% per unit	
Cost of machine		20 lakh
Cost of labour		2000/unit
Cost of raw material		Half of the labour cost/unit
Overhead Cost		20% of labour cost/ unit

If the annual demand is 2200, the company should make or buy.

3.a Discuss factors which should be considered while deciding location of the firm aboard. 5

3.b The demand for two commodities are given by: 2

$$X_1 = P_1^{-1.2}P_2^{0.5} \text{ and } X_2 = P_1^{0.3}P_2^{0.8}$$

Find out whether the two commodities X_1 and X_2 are competitive or complementary.

P.T.O.

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3.c A concrete aggregate mix must contain at least 31% sand by volume for proper batching. One source of material, which has 25% sand and 75% coarse aggregate, sells for Rs 3 per cubic meter (m^3). Another source, which has 40% sand and 60% coarse aggregate, sells for Rs. 4.40 per cubic meter (m^3). Determine the least cost per cubic meter of blended aggregates. 3

4.a Recently Rs 1000 and Rs.500 notes have been banned in India and not a legal tender any more. Discuss its implication on the Indian economy? 5

4.b A company must decide whether to buy machine A or Machine B: 5

	Machine A	Machine B
Initial Cost	Rs. 4.00,000	Rs. 8,00,000
Useful Life, in years	4	4
Salvage value at the end of machine life	Rs. 2,00,000	Rs. 5,50,000
Annual Maintenance Cost	Rs. 40,000	0

At 12 % interest rate, which machine should be selected? (Use Future Worth Method of Comparison)

5.a What do you mean by an Entrepreneur? How it is different from other factors of production? What should be done to enhance entrepreneurship among engineering students? 5

5.b A firm, Prime Manufacturing is planning to expand its production operation. It has identified three machines which are technologically capable to serve the purpose. The initial outlay and annual revenues with each of the machines are given below: 5

	Initial Outlay (Rs.)	Annual revenue	Life (in years)
Machine I	Rs. 5,45,000	Rs. 2,50,000	15
Machine II	Rs. 6,14,000	Rs. 3,30,000	12
Machine III	Rs. 6,00,900	Rs. 3,50,000	10

If the rate of interest is 12%, which machine the company should opt for? Find out the result by Annual Equivalence Method.

a Discuss those factors which are crucial for making an innovation successful business proposition. 5

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Q. 2 A laboratory prismatic slender cantilever
700 mm, has the free end angle section
Determine the horizontal and vertical
and the longitudinal stress at the top
a load of 15 N at the free end along

3

6.b A small business with an initial outlay of Rs. 12,000 yields Rs. 10,000 during the first year of its operation and the yield increases by Rs. 1,000 from its second year of operation up to its 10th year which is also the last year of operation. The salvage value is zero. Find the present worth of the business by assuming as interest rate of 18%, compounded annually. 5

7.a Can Perfect Competitions exist in the real world? Discuss giving suitable examples. 5

7.b Two years ago, a machine was purchased at a cost of Rs. 4,00,000 to be useful for ten years. Its salvage value at the end of its life is Rs. 60,000. The annual maintenance cost is Rs. 60,000. The market value of the present machine is Rs. 320000. Now a new machine to cater to the need of the present machine is available at Rs. 3,80,000 to be useful for eight years. Its annual maintenance cost is Rs. 45000. The salvage value of the new machine is Rs. 40000. Using an interest rate of 11%, find whether it is worth replacing the present machine with the new machine? 5

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